

Cargills Ceylon PLC- Optimal Routing Analysis

1. Assumptions

- Truck Capacity -For the first scenario, it is assumed that the truck has no capacity limit and can carry an unlimited load..
- Limited Capacity - In the second scenario, the truck has a limited capacity and can carry up to 5000 kg per trip.

- The distribution center is based in Katana, within the Gampaha District, and serves as the starting point for all routes.
- Traffic conditions and any other potential delays are not factored into the analysis.

2. Formulations

2.1 Optimal Route with Unlimited Truck Capacity

Z - the total distance traveled.

d_{ij} - the distance between locations i and j .

x_{ij} - binary decision variable (1 if the route from i to j is taken, 0 otherwise).

Objective Function:

Minimize the total distance traveled

$$\text{Minimize } Z = \sum_{i=0}^n \sum_{j=0, j \neq i}^n d_{ij} x_{ij}$$

Constraints-

1. Each location must be visited exactly once

$$\sum_{j=0, j \neq i}^n x_{ij} = 1, \quad \forall i \in \{1, 2, \dots, n\}$$

2. The truck returns to the starting point

$$\sum_{j=0, j \neq 0}^n x_{0j} = 1$$

2.2 Optimal Route with Limited Truck Capacity (5000 kg)

Objective Function-

The objective remains the same, to minimize the total distance traveled

$$\text{Minimize } Z = \sum_{i=0}^n \sum_{j=0, j \neq i}^n d_{ij} x_{ij}$$

Constraints-

1. Truck Capacity Constraints- Truck Capacity Constraints - Ensure that the total load does not surpass the truck's capacity.

$$\sum_{i=1}^n L_i y_i \leq 5000$$

Where y_i is a binary variable (1 if the load at location i is included in the route, 0 otherwise).

2. Each location must still be visited exactly once

$$\sum_{j=0, j \neq i}^n x_{ij} = 1, \quad \forall i \in \{1, 2, \dots, n\}$$

3. The truck returns to the starting point

$$\sum_{j=0, j \neq 0}^n x_{0j} = 1$$

3. Discussion

Unlimited Truck Capacity

When the truck has unlimited capacity, it can visit all Food City locations in one trip following the optimal route. This situation mirrors the classic Traveling Salesman Problem (TSP), where the truck completes a single continuous trip without returning to the warehouse after each stop. The main objective is to minimize the total distance traveled while ensuring each location is visited exactly once.

Limited Truck Capacity

When the truck is limited to carrying 5000 kg per trip, the problem becomes more complex. In this case, multiple trips may be necessary to meet the demand at each Food City location. The optimal solution must minimize the total distance traveled while ensuring that the demand at each store is fulfilled, possibly over several trips. This results in a longer overall travel distance due to the need for multiple trips to transport the required loads.